

Capstone Project Proposal

Physics Quest: A 2.5 D Game-Based Learning and Assessment System for Physics Education

Submitted by

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ABSTRACT

Physics Quest is a 2.5D role-playing game (RPG) built in Godot 4 that integrates game-based learning and assessment across five core physics domains: Mechanics, Electricity, Magnetism, Optics, and Thermodynamics. Each domain is represented as a spatially distinct 2.5D zone, a three-dimensional environment inhabited by 2D sprite-based characters, themed around a real-world setting such as a factory, power plant, or observatory. Players navigate these zones, interact with NPC characters modelled on historical physicists, and complete domain-specific quiz challenges drawn from a Bloom's Taxonomy-aligned question bank. Passing each quiz awards a domain badge; collecting all five badges unlocks the Hall of Physics Legends, a comprehensive final assessment stage. The system addresses documented shortcomings in conventional physics instruction: low student engagement, rote-memorization bias, absence of immediate feedback, and decontextualized concept delivery. Implemented in C# with JSON-based question storage and a local save system, Physics Quest is designed to run on standard school hardware with minimal resource requirements.

Keywords: Game-Based Learning, Physics Education, 2.5D RPG, Bloom's Taxonomy, Gamification, Assessment

1. INTRODUCTION

Physics education at the secondary level is consistently challenged by low student engagement, the delivery of abstract content, and assessment instruments that prioritize rote recall over conceptual understanding. Digital game-based learning (GBL) has emerged as a pedagogically validated alternative, one that situates knowledge within meaningful contexts, provides immediate formative feedback, and uses reward mechanics to sustain motivation [5]. Among game genres, the role-playing game (RPG) is especially well-suited to educational deployment because it combines world exploration, narrative progression, and character development into a framework that mirrors the gradual accumulation of knowledge.

The 2.5D visual style, featuring three-dimensional environments populated by 2D sprite characters, offers a middle ground between the simplicity of flat 2D assets and the immersive depth of full 3D. This style has been widely adopted in commercial games (e.g., *Cult of the Lamb*, *Paper Mario*) and is achievable in Godot 4 without the hardware demands of a full 3D pipeline, making it well-suited for deployment on school computers with limited resources.

Physics Quest combines these elements into a unified learning system: a 2.5D RPG world with five thematically grounded zones, each housing a domain-specific quiz challenge aligned to Bloom's Taxonomy. The project targets secondary and higher-secondary students, and its dual purpose is educational enrichment and formal assessment.

1.1 Problem Statement

Online and classroom physics instruction relies heavily on static lecture materials and paper-based tests. These modalities fail to provide the interactive, immediate, and contextualized learning experiences that contemporary learners expect. Instructors lack tools that simultaneously maintain engagement, deliver contextualized physics content, and produce assessable performance data. Existing educational games either focus on a single physics domain, require expensive hardware, or lack structured assessment components [1, 2]. Physics Quest is designed to address this gap.

1.2 Objectives

The main objectives of this project are:

1. To design and implement a 2.5D RPG game environment in Godot 4, covering five physics domains representing Mechanics, Electricity, Magnetism, Optics, and Thermodynamics, each with interactive NPC guidance and themed environments.
2. To develop a Bloom's Taxonomy-aligned question bank and quiz system that provides immediate

explanatory feedback and enforces a domain competency threshold.

3. To implement a badge progression and save-data system that tracks player scores, completed domains, and unlocks the final Hall of Physics Legends stage after all five badges are earned.
4. To conduct usability testing with students and measure learning outcomes using quiz performance, completion rates, and engagement feedback collected during the classroom pilot study.

2. RELATED WORK

Prior work in game-based physics education is reviewed across two categories: existing project implementations and empirical research articles. Both tables identify design features and gaps that Physics Quest directly addresses through its 2.5D multi-domain RPG architecture.

2.1 Project-Based Related Work

Table 1 surveys four notable game-based systems that have been developed and deployed for physics learning.

Table 1: Comparison of Existing Educational Game Projects

Ref.	Project	Author(s)	Description	Key Findings	Limitations
[1]	Physics Playground	Shute et al. (2021)	2D physics puzzle game with stealth assessment via evidence-centered design (ECD)	Newtonian physics understanding improves; adaptive level sequencing outperforms linear play	No narrative RPG structure; single-physics-domain scope only
[2]	PhyLab (VR Lab)	Korlat et al. (2024)	Game-based VR laboratory covering three physics topics; includes per-unit quizzes	Improved student performance and motivation in secondary school physics	Requires VR hardware; not accessible on standard school computers
[3]	Kerbal Space Program (KSP)	Squad / Mallory (2019)	3D aerospace simulation where players build and launch rockets under realistic orbital physics	Players develop physics intuition, trial-and-error problem solving and systems thinking	Single domain (aerospace); no structured assessment or curriculum alignment
[4]	PhyGame	Katanosaka et al. (2024)	Gamified online physics platform with simulation, MCQ, and badge modes	Strong engagement outcomes; badge system validated learner progression	Web-based only; lacks world-building or narrative that sustains long-term play

Across all four projects, a consistent gap is evident: none combines multi-domain coverage, an RPG narrative structure, a 2.5D visual presentation, and a curriculum-mapped badge-based assessment system within a single lightweight application. Physics Quest is designed to address this intersection.

2.2 Article-Based Related Work

Table 2 summarizes four peer-reviewed empirical studies relevant to the design and evaluation of game-based physics learning systems.

Table 2: Summary of Relevant Research Articles

Ref.	Author(s)	Study Focus	Key Findings	Limitations
[5]	Pranata (2024)	PhET simulations used as GBL tools in teaching equilibrium; quasi-experimental design	Significant gains in understanding and motivation over traditional instruction; positive student perception	Quiz-centric with no world-exploration; no multi-domain coverage
[6]	Adipat et al. (2021)	Theoretical framework for GBL; identifies engagement, problem-solving, and narrative-centered learning theories	Establishes that collaborative, story-driven, project-based activities maximize student engagement	Review paper only; no empirical game prototype built or tested
[7]	Zeng et al. (2020)	Evaluates interactive GBL in physics domain; examines impact of feedback mechanics and scoring	Structured game mechanics with immediate feedback drive measurable performance gains	No persistent save or progressive badge system; single-session scope
[8]	Rosenthal & Ratan (2022)	Communication mediation model applied to KSP; 241 undergraduates, 4-hour session	Game progress positively predicted science knowledge acquisition ($\beta = 0.21$) and gaming continuance	Only aerospace physics; no curriculum-mapped quiz assessment component

The reviewed articles collectively confirm that game-based approaches yield superior engagement and learning outcomes compared to traditional instruction, that immediate feedback is critical to formative gain, and that badge-based progression systems are empirically validated motivators. Physics Quest synthesizes these evidence-based design principles into a single, accessible 2.5D platform.

3. METHODOLOGY

The project follows an iterative, game-design-driven development methodology structured across four sequential phases. Each phase concludes with a defined testable milestone, ensuring continuous validation of both technical and educational components.

3.1 Development Phases

Phase 1: Foundation (Weeks 1–3): Establishes the Godot 4 project structure, configures three core autoload scripts (`GameManager`, `SceneManager`, `BadgeManager`), and creates placeholder scenes for all five zones. GitHub is initialized for version control. *Deliverable:* a bootable project with working scene navigation and badge state tracking.

Phase 2: Core Systems (Weeks 4–6): Implements the player controller for 2.5D movement, the JSON-based question loader, the `QuizManager` logic, and the quiz panel UI. Zone trigger volumes are wired to launch the correct domain quiz on player entry. *Deliverable:* a playable single-zone prototype with a functional quiz and badge award.

Phase 3: Content and World-Building (Weeks 7–9): Builds all five 2.5D zone scenes (Factory, Power Plant, Research Lab, Observatory, Industrial Facility), populates the question banks (15 questions per domain; 75 total, spanning four Bloom’s levels), implements NPC dialogue trees, and constructs the Hall of Physics Legends final stage and HUD. *Deliverable:* a complete five-zone game with all content.

Phase 4: Polish and Testing (Weeks 10–12): Integrates zone-specific background music and sound effects, adds player walk-cycle animations and scene transition effects, and builds the main menu. A classroom pilot with a sample student group is conducted, and feedback is used to finalize the build. *Deliverable:* a release-ready build submitted as the capstone product.

3.2 Methodology Workflow Diagram

Figure 1 presents the development workflow diagram for Physics Quest.

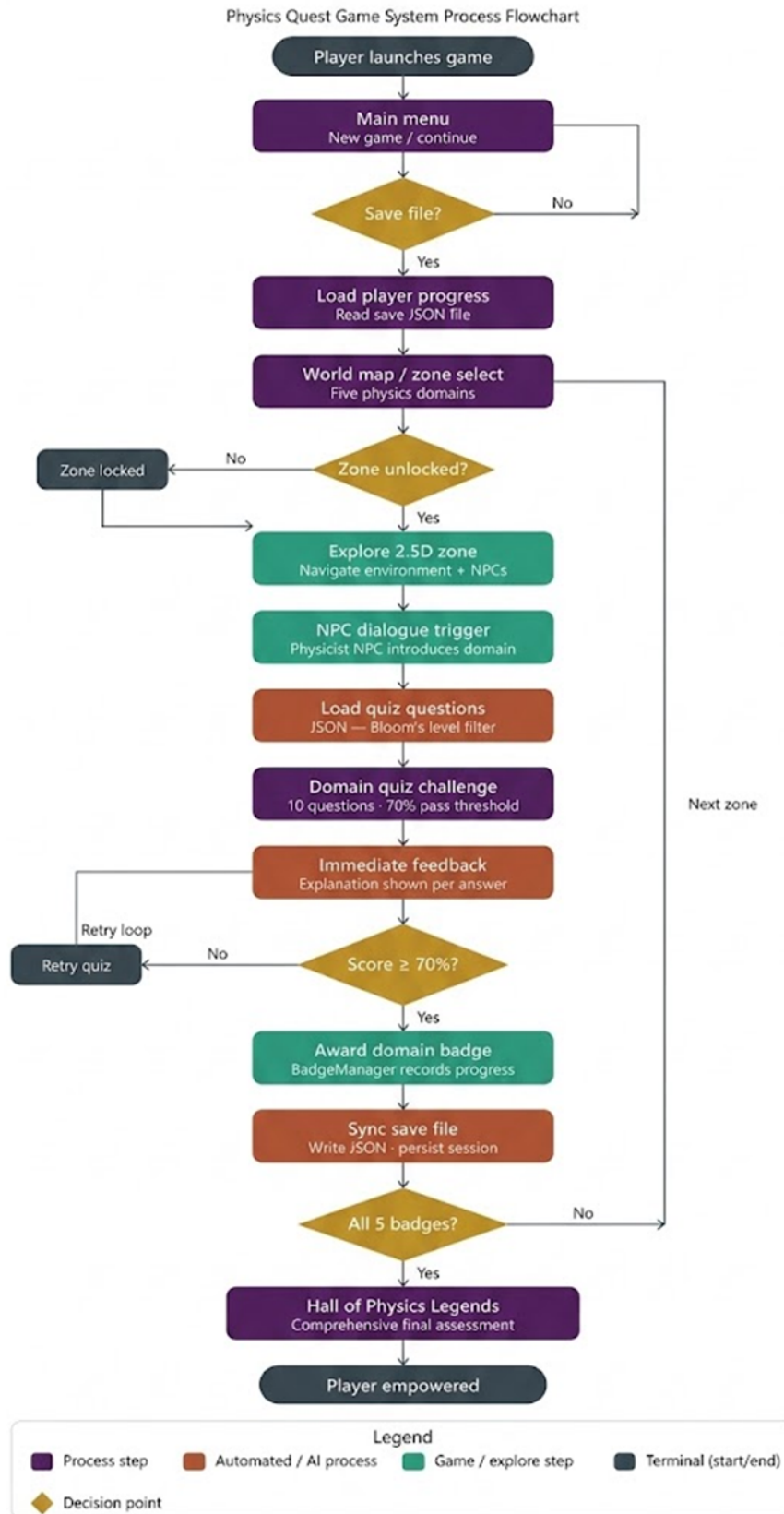


Figure 1: Development workflow diagram of the Physics Quest 2.5D game-based learning system

4. TOOLS AND TECHNOLOGIES

Table 3 lists the technology stack selected for Physics Quest. Each component was chosen based on performance, cost, educational accessibility, and compatibility with the 2.5D development approach in Godot 4.

Table 3: Tools and Technology Stack

Component	Technology	Justification
Game Engine	Godot 4 (Standard / C#)	Lightweight (~300 MB RAM idle); open-source; excellent 2.5D support via 3D world + 2D sprites
Scripting Language	C#	Python-like syntax; tightly integrated with Godot; no compilation step required
2.5D Rendering	Godot 4 SubViewport + 3D Scene	3D environment with 2D sprite characters achieves 2.5D visual style natively
Quiz / Question Data	JSON files	Human-readable; editable by educators without database or programming knowledge
Save System	JSON via <code>user://</code> path	Persistent, cross-session progress; no external database or internet required
Code Editor	VS Code + Godot Tools extension	Superior autocomplete, linting and debugging over Godot's built-in editor
Sprite / Art	Piskel / Krita / Blender	Free tools for 2D sprites (Piskel/Krita) and low-poly 3D zone assets (Blender)
Version Control	GitHub	Team collaboration, commit history, and branch-based iterative development
Audio	Royalty-free + Godot AudioStreamPlayer	In-engine audio; no external library required

5. EXPECTED RESULTS / OUTCOMES

The following outcomes are expected upon project completion:

- A fully playable 2.5D RPG game with five physics-domain zones, running on standard school hardware under Godot 4.
- A structured, Bloom's Taxonomy-aligned question bank comprising 75 verified physics questions across five domains.
- A badge-based assessment system that records per-domain competency and unlocks a final comprehensive stage upon completion of all five domains.
- Performance data from a classroom pilot study, including accuracy rates per Bloom's level, session completion rates, and student-reported engagement scores.
- A replicable game-based assessment framework that can be extended to other STEM subjects with minimal architectural modification.

6. Evaluation Method

The evaluation of Physics Quest is conducted across two dimensions: technical performance and educational effectiveness. Both are assessed during Phase 4 of development (Weeks 10–12) through a combination of internal testing and a controlled classroom pilot.

6.1 Technical Evaluation

The system is evaluated against the following functional criteria prior to the classroom pilot:

Criterion	Metric	Target
Scene load time	Time (seconds)	≤ 3 s per zone transition
Quiz accuracy	Correct badge awarded per domain	100% correctness across 5 domains
Save/load integrity	Progress retained across sessions	No data loss across 10 restart tests
RAM usage (idle)	Memory consumption	≤ 350 MB on standard school hardware
Crash rate	Playthroughs without runtime error	$\geq 95\%$ error-free sessions

6.2 Educational Effectiveness Evaluation

A classroom pilot is conducted with a sample group of secondary students. The evaluation uses a pre-test / post-test design to measure conceptual gain:

- **Pre-test:** A 20-question multiple-choice paper covering the five physics domains is administered before gameplay.
- **Intervention:** Students play Physics Quest for one full session (approximately 60–90 minutes), completing at least two domain zones.
- **Post-test:** The same 20-question instrument is re-administered immediately after gameplay.

Conceptual gain is measured using Hake's Normalised Gain formula:

$$g = \frac{\text{Post-test \%} - \text{Pre-test \%}}{100 - \text{Pre-test \%}} \quad (1)$$

A gain of $g \geq 0.3$ is considered acceptable; $g \geq 0.7$ is classified as high gain [9].

6.3 Engagement and Usability Evaluation

Following the gameplay session, students complete a short-structured survey based on the System Usability Scale (SUS) [10] and a five-item Likert engagement questionnaire assessing:

- Motivation to complete remaining zones
- Perceived educational value of in-game quizzes
- Clarity of NPC dialogue and zone context
- Satisfaction with immediate post-answer feedback
- Likelihood of recommending the game to peers

Results are reported as mean scores with standard deviation. A SUS score of ≥ 70 is the target threshold for acceptable usability.

6.4 Per-Domain Performance Analysis

In-game quiz data is logged per player session, capturing accuracy per Bloom's level (Remember, Understand, Apply, Analyze) across all five domains. This allows identification of which Bloom's levels and which domains present the greatest difficulty, informing future question bank refinements.

7. WORK PLAN

Figure 2 presents the twelve-week project Gantt chart. Filled cells indicate the weeks in which each deliverable is actively under development.

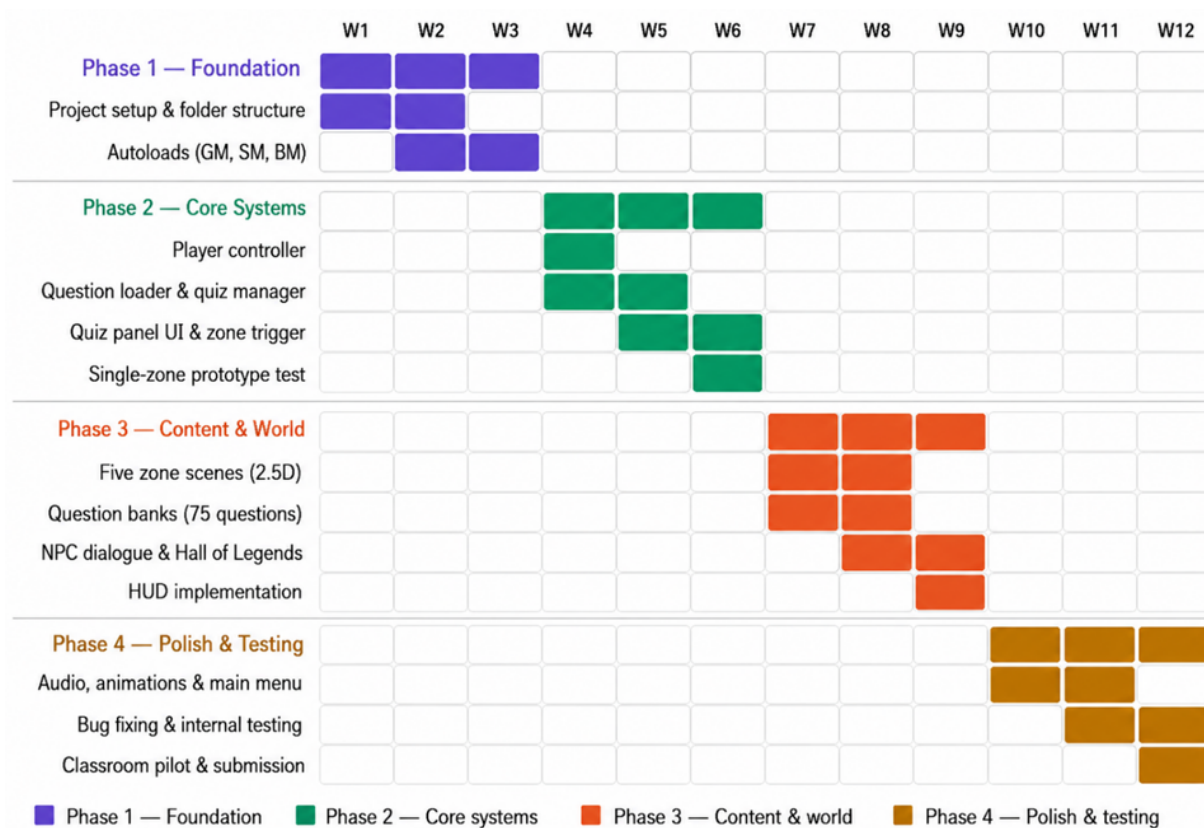


Figure 2: Project Timeline

8. CONCLUSION

Physics Quest presents a rigorously designed 2.5D game-based learning and assessment system that addresses four well-documented failures of conventional physics instruction, which solves problems such as low engagement, rote-memorization bias, delayed feedback, and decontextualized content, by embedding Bloom's Taxonomy-aligned physics quizzes within an immersive, multi-zone RPG built in Godot 4. The proposed system draws on a substantive body of peer-reviewed literature confirming that game-based approaches, badge-driven progression, and immediate explanatory feedback each independently improve learner outcomes in STEM domains; Physics Quest combines all three mechanisms into a single, hardware-accessible platform. The structured four-phase development methodology, verified technology stack, and twelve-week timeline provide a realistic and testable path to a release-ready capstone product, with the classroom pilot study in Phase 4 producing empirical performance data that will validate the framework's educational effectiveness and inform future extensions to additional STEM subjects.

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